Math **163** Name (PLEASE PRINT) Srinarayan Srikanth

Exam **2A** **2026** Fall/Spring/Summer I /**Summer II** Score 0

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	3	3								6	
1b	3	3								6	
1c	1	1	1	2						5	
2a	6									6	
2b	6									6	
2c	5									5	
3a	2	4								6	
3b	6									6	
3c	1	1	1	2						5	
								Total		51	

You may discuss and work with anyone

Instructions

- All work must be done using a **MECHANICAL PENCIL following instructions given**.
- If you have any questions at all, please **ask me**. This includes the following possibilities:
    - something is **illegible,** or **ambiguous**, or **unclear** to you.
- **You may use your homework, notes, books, and calculators unless otherwise stated during the exam.**
- **You will not get any partial credit**. You have to show **absolutely all your work** on all questions, **exactly as in the notes, with no changes, variations, or omissions in form, unless otherwise noted whether orally or in writing**. If your answer is correct, but your procedure is incorrect or is not allowed by the instructions, you will get no credit all. All work must be displayed using **correct syntax**, and all **explanations**, written in **complete sentences**, must be complete.
- **You must write neatly, clearly and legibly. You must draw every straight-line segment other than those in the list: (**$=, \neq, \equiv, <, \leq, >, \geq, +, -, \times, \div, \ \neg, \vee, \wedge, \|, \perp, \setminus, \rightarrow, \leftrightarrow, \mapsto$**) with your ruler. At least 1 point will be taken off for each instance of violation. Signs of equality or equivalence must line up, one below the other on the left of expressions. Every symbol has a specific meaning and may be used only in this sense and in no other. Syntax defined in the notes must be followed exactly to the letter.**
- Remember that 'does not make sense' is a possible answer.
- I can grade you only on what you **actually** put on paper; I cannot read minds. You cannot say later that what you had written down is not what you had meant. **You need to learn to write exactly what you mean and mean exactly what you write**.
- **The answers for a problem should be noted on the page on which the problem appears in the areas indicated. All work for the problem should be shown on sheets that follow the page on which the problem appears numbered as follows. Work for a problem that occurs on page n should be on blank (not ruled) sheets after page n, numbered, n0, n1, n2 etc. on the lower right-hand corner as in this document. For example, work for a problem on page 502 should be on pages numbered 5020, 5021, 5022 etc. as needed. Work that corresponds to different parts must be separated from each other by straight lines drawn with a ruler parallel to the edges of the paper and labelled correctly as in the Pages from Student Work as found under Files on CANVAS.**
- **You should submit a SINGLE document.**
- **You may NOT include pages that you have NOT attempted**.
- **As shown in the class means as shown in the notes.**

You may discuss and work with anyone 401

Instructions:

Your J-number = 00499317

$a := mnm_{k \in 1..8}\{J_k | (k \in 1..8) \wedge (J_k \neq 0)\} = 1$

$b := mxm(J_k | k \in 1..8) = 9$

$c := \left\lfloor \left(\frac{b+a}{2} \right) \right. = \left\lfloor \frac{9+1}{2} \right\rfloor = \left\lfloor \frac{10}{2} \right\rfloor = \lfloor 5 \rfloor = 5 =$

$d :=$ the smallest **odd** number **strictly greater** than c = 7

R Do not use decimals anywhere.

R Do not make any extraneous marks anywhere on the exam.

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in any set in the above. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work.

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You will lose points for otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

R On (1a) use the binomial or the multinomial theorem as noted. Express your answer as a polynomial with **integer coefficients**, **ordering the unknowns within each summand numero-alphabetically, writing the summands VERTICALLY, and for every summand, ordering the summands lexicographically exactly as shown in class. Note that(-1) is the only allowed negative number**. Show calculations for all coefficients that occur on pages that follow and note only the answer on the page containing the problem.

6 (1a) Expand using the binomial theorem **exactly as in the Notes.**

3 (i) $(b - ax)^c$

My problem: $\left((9) - (1)x\right)^{(5)}$

$= \left((9) + (-1)x\right)^{(5)}$

$$
\begin{aligned}
= \quad & 59049 \\
& + (-1)32805x \\
& + 7290x^2 \\
& + (-1)810x^3 \\
& + 45x^4 \\
& + (-1)x^5
\end{aligned}
$$

1a.i.0

$$\frac{(a+b)^{n} = \sum^{n} \left(\binom{n}{k}\right) a^{n-k} b^{k}}{(9+(-1)x)^{5} = \sum \left(\binom{5}{k}\right) (9)^{5-k} ((-1)x)^{k}} \left\langle \begin{matrix} n \leftarrow 5 \\ a \leftarrow 9 \\ b \leftarrow (-1)x \end{matrix} \right\rangle$$

1a.i.1

$$\begin{aligned} & \sum^{5} \left(\binom{5}{k}\right) (9)^{5-k} ((-1)x)^{k} \\ = \; & \left(\binom{5}{0}\right) (9)^{5} ((-1)x)^{0} \\ + \; & \left(\binom{5}{1}\right) (9)^{4} ((-1)x)^{1} \\ + \; & \left(\binom{5}{2}\right) (9)^{3} ((-1)x)^{2} \\ + \; & \left(\binom{5}{3}\right) (9)^{2} ((-1)x)^{3} \\ + \; & \left(\binom{5}{4}\right) (9)^{1} ((-1)x)^{4} \\ + \; & \left(\binom{5}{5}\right) (9)^{0} ((-1)x)^{5} \end{aligned}$$

1a.i.2

$$\begin{aligned} & \left(\binom{5}{0}\right) \\ = \; & 1 \\ & \left(\binom{5}{0}\right) (9)^{5} ((-1)x)^{0} \\ = \; & 59049 \end{aligned}$$

1a.i.3

$$\begin{aligned} & \left(\binom{5}{1}\right) \\ = \; & 5 \\ & \left(\binom{5}{1}\right) (9)^{4} ((-1)x)^{1} \\ = \; & (-1)32805x \end{aligned}$$

1a.i.4

$$
\begin{aligned}
& \left(\binom{5}{2}\right) \\
= \ & \frac{(5)!}{(2)!(3)!} \\
= \ & \frac{(5)(4)\ \cancel{(3)}\ \cancel{(2)}\ \cancel{(1)}}{\cancel{(3)}\ \cancel{(2)}\ \cancel{(1)}} \\
= \ & \frac{(5)(4)}{(2)(1)} \\
= \ & \frac{20}{2} \\
= \ & 10 \\
& \left(\binom{5}{2}\right)(9)^{3}((-1)x)^{2} \\
= \ & 7290xx \\
= \ & 7290x^{2}
\end{aligned}
$$

1a.i.5

$$
\begin{aligned}
& \left(\binom{5}{3}\right) \\
= \ & \left(\binom{5}{5-3}\right) \\
= \ & \left(\binom{5}{2}\right) \\
= \ & 10 \\
& \left(\binom{5}{3}\right)(9)^{2}((-1)x)^{3} \\
= \ & (-1)810xxx \\
= \ & (-1)810x^{3}
\end{aligned}
$$

1a.i.6

$$
\begin{aligned}
& \left(\binom{5}{4}\right) \\
= \ & \left(\binom{5}{5-4}\right) \\
= \ & \left(\binom{5}{1}\right) \\
= \ & 5 \\
& \left(\binom{5}{4}\right)(9)^{1}((-1)x)^{4} \\
= \ & 45xxxx \\
= \ & 45x^{4}
\end{aligned}
$$

$$
\begin{aligned}
& \left(\binom{5}{5}\right) \\
= \ & \left(\binom{5}{5-5}\right) \\
= \ & \left(\binom{5}{0}\right) \\
= \ & 1 \\
& \left(\binom{5}{5}\right)(9)^{0}((-1)x)^{5} \\
= \ & (-1)xxxxx \\
= \ & (-1)x^{5}
\end{aligned}
$$

(1a) Expand using the multinomial theorem **exactly as in the Notes.**

3 (ii) $(c - x + y)^{1+a}$

My problem: $\left((5) - (1)x + y\right)^{1+(1)}$

$$= \left((5) + (-1)x + y\right)^{(2)}$$

$$= \begin{array}{l} 25 \\ \left| \begin{array}{l} +(-1)10x \\ +x^2 \\ +(-1)2xy \\ +10y \\ +y^2 \end{array} \right. \end{array}$$

1a.ii.0

$$\frac{(a+b+c)^{n} = \sum (n,k_1,k_2,k_3)\, a^{k_1} b^{k_2} c^{k_3}}{(5+(-1)x+y)^{2} = \sum (2,k_1,k_2,k_3)\, (5)^{k_1} ((-1)x)^{k_2} (y)^{k_3}} \left\langle \begin{matrix} n \longleftarrow 2 \\ a \longleftarrow 5 \\ b \longleftarrow (-1)x \\ c \longleftarrow y \end{matrix} \right\rangle$$

1a.ii.1

$$\begin{aligned} & 2 \\ = \; & 2+0+0 \\ = \; & 0+2+0 \\ = \; & 0+0+2 \\ = \; & 1+1+0 \\ = \; & 1+0+1 \\ = \; & 0+1+1 \end{aligned}$$

1a.ii.2

$$\begin{aligned} & \sum(2,k_1,k_2,k_3)\, (5)^{k_1} ((-1)x)^{k_2} (y)^{k_3} \\ = \; & (2,2,0,0)\, (5)^{2} ((-1)x)^{0} (y)^{0} \\ + \; & (2,0,2,0)\, (5)^{0} ((-1)x)^{2} (y)^{0} \\ + \; & (2,0,0,2)\, (5)^{0} ((-1)x)^{0} (y)^{2} \\ + \; & (2,1,1,0)\, (5)^{1} ((-1)x)^{1} (y)^{0} \\ + \; & (2,1,0,1)\, (5)^{1} ((-1)x)^{0} (y)^{1} \\ + \; & (2,0,1,1)\, (5)^{0} ((-1)x)^{1} (y)^{1} \end{aligned}$$

1a.ii.3

$$\begin{aligned} & (2,2,0,0) \\ = \; & \frac{(2)!}{(2)!(0)!(0)!} \\ = \; & 1 \\ & (2,2,0,0)\, (5)^{2} ((-1)x)^{0} (y)^{0} \\ = \; & 25 \end{aligned}$$

1a.ii.4

$$\begin{aligned} & (2,0,2,0) \\ = \; & \frac{(2)!}{(0)!(2)!(0)!} \\ = \; & 1 \\ & (2,0,2,0)\, (5)^{0} ((-1)x)^{2} (y)^{0} \\ = \; & xx \\ = \; & x^2 \end{aligned}$$

1a.ii.5

$$
\begin{aligned}
& (2,0,0,2) \\
= & \frac{(2)!}{(0)!(0)!(2)!} \\
= & 1 \\
& (2,0,0,2)\ (5)^{\ 0}\ ((-1)x)^{\ 0}\ (y)^{\ 2} \\
= & yy \\
= & y^2
\end{aligned}
$$

1a.ii.6

$$
\begin{aligned}
& (2,1,1,0) \\
= & \frac{(2)!}{(1)!(1)!(0)!} \\
= & 2 \\
& (2,1,1,0)\ (5)^{\ 1}\ ((-1)x)^{\ 1}\ (y)^{\ 0} \\
= & (-1)10x
\end{aligned}
$$

1a.ii.7

$$
\begin{aligned}
& (2,1,0,1) \\
= & \frac{(2)!}{(1)!(0)!(1)!} \\
= & 2 \\
& (2,1,0,1)\ (5)^{\ 1}\ ((-1)x)^{\ 0}\ (y)^{\ 1} \\
= & 10y
\end{aligned}
$$

1a.ii.8

$$
\begin{aligned}
& (2,0,1,1) \\
= & \frac{(2)!}{(0)!(1)!(1)!} \\
= & 2 \\
& (2,0,1,1)\ (5)^{\ 0}\ ((-1)x)^{\ 1}\ (y)^{\ 1} \\
= & (-1)2xy
\end{aligned}
$$

6 (1b) Solve the following recurrence relations with the indicated initial conditions, using the formula for second-order recurrences in the Notes.

$$\rho(0) \coloneqq 1$$
$$\rho(1) \coloneqq 2$$

3 (i) $$\begin{cases} \forall n \in \mathbb{N} \quad \Big(\rho(n+2)\Big) - (b+c)\Big(\rho(n+1)\Big) + (bc)\Big(\rho(n)\Big) = 0 \end{cases}$$

First rewrite the recurrence with your values of ***b*** **and** ***c***.

My problem:

$$\Big(\rho(n+2)\Big) - \Big((9)+(5)\Big)\Big(\rho(n+1)\Big) + \Big((9)(5)\Big)\Big(\rho(n)\Big) = 0$$

My problem:

$$\Big(1\Big)\Big(\rho(n+2)\Big) + \Big((-14)\Big)\Big(\rho(n+1)\Big) + (45)\Big(\rho(n)\Big) = 0$$

My solution: $\forall n \in \mathbb{N}$

$$\rho(n)$$

$$= \left(\frac{7}{4}\right)(5)^n - \left(\frac{3}{4}\right)(9)^n$$

1b.i.0

$$\rho(n{+}2) - 14\rho(n{+}1) + 45\rho(n) = 0$$

$$\frac{-(-14\rho(n{+}1){+}45\rho(n)) \qquad\qquad -(-14\rho(n{+}1){+}45\rho(n))}{(1)(\rho(n{+}2)) = -(-14)(\rho(n{+}1)) - (45)(\rho(n))}$$

1b.i.1

$$\frac{\frac{\frac{\frac{\frac{\frac{(1)n^2{+}(-14)n{+}45 = 0}{(n{-}9)(n{-}5) = 0}}{(n{-}9{=}0) \vee (n{-}5{=}0)}}{(n{=}9) \vee (n{=}5)}}{(r_1{=}5) \wedge (r_2{=}9)}}{r_1 \neq r_2}}{\forall n \in \mathbb{N}\ \rho(n) = k_1\,(5)^n + k_2(9)^n}$$

1b.i.2

$$\frac{\frac{\rho(n) = k_1(5)^n + k_2(9)^n}{\rho(0) = k_1(5)^0 + k_2(9)^0}\ \langle n \longleftarrow 0 \rangle}{\rho(0) = k_1(1) + k_2(1)} \qquad \frac{\frac{\rho(n) = k_1(5)^n + k_2(9)^n}{\rho(1) = k_1(5)^1 + k_2(9)^1}\ \langle n \longleftarrow 1 \rangle}{\rho(1) = k_1(5) + k_2(9)}$$

1b.i.3

$$\frac{1 = k_1 + k_2}{\frac{-k_1 \qquad\qquad -k_1}{k_2 = 1 - k_1}} \qquad \frac{\frac{\frac{2 = k_1(5) + k_2(9)}{2 = k_1(5) + (1{-}k_1)(9)}}{2 = 5k_1 + 9 - 9k_1}}{2 = 9 - 4k_1}$$

1b.i.4

$$\frac{\frac{4k_1 = 7}{\div 4 \qquad\quad \div 4}}{k_1 = \frac{7}{4}} \qquad \begin{aligned} k_2 &= 1 - k_1 \\ &= 1 - \frac{7}{4} \\ &= \frac{-3}{4} \end{aligned}$$

1b.i.5

$$\begin{aligned} &\forall n \in \mathbb{N}\ \rho(n) \\ &= \left(\frac{7}{4}\right)(5)^n + \left(\frac{-3}{4}\right)(9)^n \end{aligned}$$

(1b) Continued. Solve the following recurrence relations with the indicated initial conditions, using the formula for second-order recurrences given in the Notes, using substitution. Prove by substitution that your answer is correct.

$$\rho(0) \coloneqq 1$$
$$\rho(1) \coloneqq 2$$

3 (ii) $$\left\{\forall n \in \mathbb{N} \quad (a^2)\left(\rho(n+2)\right) - (2ac)\left(\rho(n+1)\right) + (c^2)\left(\rho(n)\right) = 0\right.$$

First rewrite with your values of ***a*** **and** ***c***.

My problem:

$$\left((1)^2\right)\left(\rho(n+2)\right) - \left(2(1)(5)\right)\left(\rho(n+1)\right) + \left((5)^2\right)\left(\rho(n)\right) = 0$$

My problem:

$$\left(1\right)\left(\rho(n+2)\right) + \left((-10)\right)\left(\rho(n+1)\right) + (25)\left(\rho(n)\right) = 0$$

My solution: $\forall n \in \mathbb{N}$

$$\rho(n)$$

$$= (1)(5)^n + \left(\frac{-3}{5}\right)(n(5)^n)$$

1b.ii.0

$$(1)\rho(n+2)+(-10)\rho(n+1)+(25)\rho(n)=0$$

$$\frac{-((-10)\rho(n+1)+(25)\rho(n)) \qquad -((-10)\rho(n+1)+(25)\rho(n))}{(1)(\rho(n+2))=(10)(\rho(n+1))-(25)(\rho(n))}$$

1b.ii.1

$$\frac{\frac{\frac{\frac{(1)n^2+(-10)n+25=0}{(n-5)^2=0}}{n=5}}{r_1=n=r_2}}{\forall n\in\mathbb{N}\ \rho(n)=k_1(5)^n+k_2n(5)^n}$$

1b.ii.2

$$\frac{\rho(n)=k_1(5)^n+k_2n(5)^n}{\rho(0)=k_1(5)^0+k_2(0)(5)^0}\ \langle n\leftarrow 0\rangle \qquad \frac{\rho(n)=k_1(5)^n+k_2n(5)^n}{\rho(1)=k_1(5)^1+k_2(1)(5)^1}\ \langle n\leftarrow 1\rangle$$

$$\rho(0)=k_1(1)+k_2(0)(1) \qquad \rho(1)=k_1(5)+k_2(1)(5)$$

1b.ii.3

$$\frac{\rho(0)=k_1(1)=1}{k_1=1} \qquad \frac{2=k_1(5)+k_2(1)(5)}{\frac{2=(1)(5)+5k_2}{2=5+5k_2}}$$

1b.ii.4

$$\frac{\frac{5k_2=-3}{\div 5 \qquad \div 5}}{k_2=\frac{-3}{5}}$$

1b.ii.5

$$\forall n\in\mathbb{N}\ \rho(n)$$

$$=\ (1)(5)^n+(\tfrac{-3}{5})n(5)^n$$

1b.ii.6

$$\forall n\in\mathbb{N}\ (\ P(n)\ :=\ (\ \rho(\ n)=(1)(5)^n+(\tfrac{-3}{5})n(5)^n\ )\ )$$

1b.ii.7

$$\frac{P(n)}{P(0)}\ \langle n\leftarrow 0\rangle \qquad \frac{\rho(0)=(1)(5)^0+(-3\div 5)(0)(5)^0}{\rho(0)=1}\ \text{BC(0)}$$

1b.ii.8

$$\frac{P(n)}{P(1)}\ \langle n\leftarrow 1\rangle \qquad \frac{\rho(1)=(1)(5)^1+(-3\div 5)(1)(5)^1}{\rho(1)=2}\ \text{BC(1)}$$

1b.ii.9

RcS $\dfrac{P(0)\,P(1)\cdots P(n)\,P(n{+}1)}{\rho(n{+}2)}$

$= 10\rho(n{+}1) - 25\rho(n)$

$= 10((1)(5)^{n+1} + (\frac{-3}{5})(n{+}1)(5)^{n+1})$

$= -25((1)(5)^{n} + (\frac{-3}{5})n(5)^{n})$

$= (50 - 30(n{+}1) - 25 + 15n)(5)^{n}$

$= (-15n - 5)(5)^{n}$

$= (-3n - 1)(5)^{n+1}$

$= (1)(5)^{n+2} + (\frac{-3}{5})(n{+}2)(5)^{n+2}$

$\dfrac{}{P(n{+}2)}$

1b.ii.10

$$\frac{P(0)}{\dfrac{P(0)\,P(1)\cdots P(n)\,P(n{+}1)}{P(n{+}2)}}$$

$\forall n \in \mathbb{N}(P(n))$

5 (1c) Starting with two sets A nd B with $a \coloneqq \nu(A)$ and $b \coloneqq \nu(B)$, that is use as many of the operations on sets that you have learnt from the set:

$$\left\{(\quad)^c, \cup, \cap, \setminus, \prod \quad, \coprod \quad, Fnc(A,B), (\quad), \underset{\sim}{-}\right\}$$

(i) to **construct a set S whose cardinalities are the following**, and

(ii) to **construct word-problems whose solutions require counting the number of elements of the sets you constructed in (i)**.

<u>You will get 0, if you calculate numerical values of any of the expressions or subexpressions in (1C)</u>.

Draw a formation-tree for the expression on pages that follow, numbered as per instructions.

Draw a formation-tree for the expression-set that you have constructed on the pages that follow, numbered as per instructions.

Prove on pages that follow, properly numbered, that the cardinality of the set that you have constructed equals the given expression.

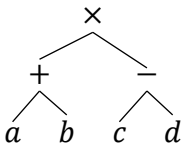
On the right side write the word problem step by step respecting the formation-tree of the expression that you have constructed exactly as shown in class.

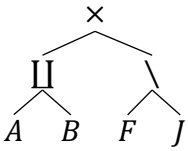
(1c) Continued

1 (i) $(a+b)(c-d)$

My expression:

$S \coloneqq (A \coprod B) \times (F \setminus J) \quad (v(F)=c, v(J)=d, J \setminus F = \{\,\})$





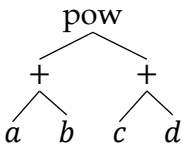
$v(A \coprod B) = v(A) + v(B) = a+b$
$v(F \setminus J) = v(F) - v(J) = c-d$
$v(S) = v((A \coprod B) \times (F \setminus J)) = v(A \coprod B) \times v(F \setminus J) = (a+b)(c-d)$

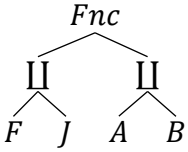
(1c) Continued

1 (ii) $\left(a+b\right)^{c+d}$

My expression:

$S \coloneqq Fnc(F\coprod J, A\coprod B)$





$\nu(F\coprod J) = c+d$, $\nu(A\coprod B) = a+b$

$\nu(Fnc(P,R)) = \nu(R)^{\nu(P)}$

$\nu(S) = \nu(A\coprod B)^{\nu(F\coprod J)} = (a+b)^{(c+d)}$

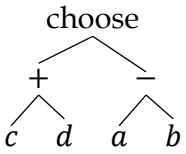
(1c) Continued

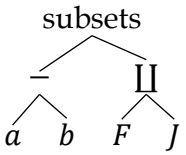
1 (iii) $\binom{c+d}{a-b}$

My expression:

$S \coloneqq \text{subsets}((a-b), F \coprod J)$

$B \setminus A = \{\}$





$v(F \coprod J) = c+d$

$v(P)=n \rightarrow v(\text{subsets}(k,P)) = \binom{n}{k}$

$v(S) = \binom{c+d}{a-b}$

(1c) Continued

2 (iv) $\begin{pmatrix} \binom{p+q}{r+s} \\ \binom{a-b}{c-d} \end{pmatrix}$

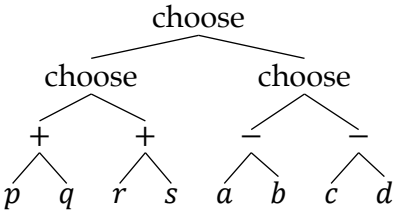
My expression:

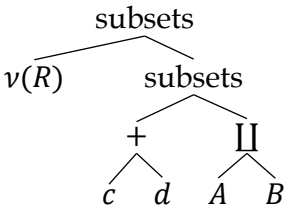
$P \coloneqq \text{subsets}((c+d), A \coprod B)$

$R \coloneqq \text{subsets}((c-d), A \setminus B)$ ($B \setminus A = \{\}$)

$S \coloneqq \text{subsets}(v(R), P)$

$p \coloneqq a, q \coloneqq b, r \coloneqq c, s \coloneqq d$





$P \coloneqq \text{subsets}((c+d), A \coprod B) \quad v(P) = \begin{pmatrix} a+b \\ c+d \end{pmatrix}$

$R \coloneqq \text{subsets}((c-d), A \setminus B) \quad v(R) = \begin{pmatrix} a-b \\ c-d \end{pmatrix}$

$v(S) = \begin{pmatrix} v(P) \\ v(R) \end{pmatrix}$

6 (2a) Consider the following recurrence relation for $\varphi(n)$ and $\alpha(n)$ and **answer the questions**:

$$\begin{cases} & \alpha(0) \coloneqq a \\ \forall n \in \mathbb{N} & \alpha(n+1) \coloneqq \big(\varphi(n)\big)b \end{cases}$$

$$\begin{cases} & \varphi(0) \coloneqq \alpha(0) + c \\ \forall n \in \mathbb{N} & \varphi(n+1) \coloneqq \big(\alpha(n)\big)d + a \end{cases}$$

Show calculations on pages that follow, numbered according to instructions.

Rewrite the recursion using your values of a, b, c, and d.

My problem:

$$\begin{cases} & \alpha(0) \coloneqq (\,1\,) \\ \forall n \in \mathbb{N} & \alpha(n+1) \coloneqq \big(\varphi(n)\big)(\,9\,) \end{cases} \qquad \begin{cases} & \varphi(0) \coloneqq \alpha(0) + c = (\,1\,) + (\,5\,) \\ \forall n \in \mathbb{N} & \varphi(n+1) \coloneqq \big(\alpha(n)\big)(\,7\,) + (\,1\,) \end{cases}$$

Show full computations on pages that follow, numbered as per instructions, for all values and your conjecture:

$\alpha(0) = 1$ $\varphi(0) = 6$

$\alpha(1) = 54$ $\varphi(1) = 8$

$\alpha(2) = 72$ $\varphi(2) = 379$

$\alpha(3) = 3411$ $\varphi(3) = 505$

Conjecture : $\forall n \in \mathbb{N}$

$\alpha(n) =$ $\varphi(n) =$

$$\alpha(n) = (63)^{\left\lfloor \frac{n}{2} \right\rfloor}((1)\left(\frac{1+(-1)^{n}}{2}\right) + (54)\left(\frac{1+(-1)^{n+1}}{2}\right)) + (9)\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor}-1}{62}\right)$$

$$\varphi(n) = (63)^{\left\lfloor \frac{n}{2} \right\rfloor}((6)\left(\frac{1+(-1)^{n}}{2}\right) + (8)\left(\frac{1+(-1)^{n+1}}{2}\right)) + (1)\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor}-1}{62}\right)$$

2a.0

$$\begin{aligned} & \varphi(0) \\ = \ & \alpha(0)+5 \\ = \ & 1+5 \\ = \ & 6 \end{aligned}$$

2a.1

$$\frac{\alpha(n+1)=\varphi(n)(9)}{\alpha(0+1)=\varphi(0)(9)}\ \langle n \leftarrow 0 \rangle$$

$$\begin{aligned} & \alpha(1)=(6)(9) \\ = \ & 54 \end{aligned}$$

2a.2

$$\frac{\varphi(n+1)=\alpha(n)(7)+1}{\varphi(0+1)=\alpha(0)(7)+1}\ \langle n \leftarrow 0 \rangle$$

$$\begin{aligned} & \varphi(1)=(1)(7)+1 \\ = \ & 8 \end{aligned}$$

2a.3

$$\frac{\alpha(n+1)=\varphi(n)(9)}{\alpha(1+1)=\varphi(1)(9)}\ \langle n \leftarrow 1 \rangle$$

$$\begin{aligned} & \alpha(2)=(8)(9) \\ = \ & 72 \end{aligned}$$

2a.4

$$\frac{\varphi(n+1)=\alpha(n)(7)+1}{\varphi(1+1)=\alpha(1)(7)+1}\ \langle n \leftarrow 1 \rangle$$

$$\begin{aligned} & \varphi(2)=(54)(7)+1 \\ = \ & 379 \end{aligned}$$

2a.5

$$\frac{\alpha(n+1)=\varphi(n)(9)}{\alpha(2+1)=\varphi(2)(9)}\ \langle n \leftarrow 2 \rangle$$

$$\begin{aligned} & \alpha(3)=(379)(9) \\ = \ & 3411 \end{aligned}$$

2a.6

$$\frac{\varphi(n+1) = \alpha(n)(7)+1}{\varphi(2+1) = \alpha(2)(7)+1} \langle n \leftarrow 2 \rangle$$

$$\varphi(3) = (72)(7)+1$$

$$= \ 505$$

2a.7

$\alpha(n+2)$	$\varphi(n+2)$
$= \ \varphi(n+1)(9)$	$= \ \alpha(n+1)(7)+1$
$= \ (\alpha(n)(7)+1)(9)$	$= \ (\varphi(n)(9))(7)+1$
$= \ (63)\alpha(n)+9$	$= \ (63)\varphi(n)+1$

2a.8

$$m \coloneqq \left\lfloor \frac{n}{2} \right\rfloor$$

$$\alpha(2m) = (63)^{m}(1 - (9 \div 62)) + (9 \div 62)$$

$$\alpha(2m+1) = (63)^{m}(54 - (9 \div 62)) + (9 \div 62)$$

2a.9

$$\varphi(2m) = (63)^{m}(6 - (1 \div 62)) + (1 \div 62)$$

$$\varphi(2m+1) = (63)^{m}(8 - (1 \div 62)) + (1 \div 62)$$

2a.10

$$m \coloneqq \left\lfloor \frac{n}{2} \right\rfloor$$

$$\alpha(n) = (63)^{\left\lfloor \frac{n}{2} \right\rfloor}((1)\left(\frac{1+(-1)^{n}}{2}\right) + (54)\left(\frac{1+(-1)^{n+1}}{2}\right)) + (9)\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor} - 1}{62}\right)$$

$$\varphi(n) = (63)^{\left\lfloor \frac{n}{2} \right\rfloor}((6)\left(\frac{1+(-1)^{n}}{2}\right) + (8)\left(\frac{1+(-1)^{n+1}}{2}\right)) + (1)\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor} - 1}{62}\right)$$

6 (2b) Find conjectures for the sums:

$$Sm(\alpha, n) \coloneqq S(\alpha, n) \coloneqq \sum_{k=0}^{n} \alpha(k) \quad \text{and}$$

$$Sm(\varphi, n) \coloneqq S(\varphi, n) \coloneqq \sum_{k=0}^{n} \varphi(k)$$

where $\alpha(n)$ and $\varphi(n)$ are defined, as in (2a) by the recursion appearing below,

$$\begin{cases} \boldsymbol{\alpha(0) \coloneqq a} \\ \forall \boldsymbol{n} \in \mathbb{N} \quad \boldsymbol{\alpha(n+1) \coloneqq \big(\varphi(n)\big)b} \end{cases} \qquad \begin{cases} \boldsymbol{\varphi(0) \coloneqq \alpha(0) + c} \\ \forall \boldsymbol{n} \in \mathbb{N} \quad \boldsymbol{\varphi(n+1) \coloneqq \big(\alpha(n)\big)d + a} \end{cases}$$

Show computations on pages that follow, numbered according to instructions, to indicate how you arrive at your conjecture.

Rewrite the recursion using your values of a, b, and c.

My problem:

$$\begin{cases} \alpha(0) \coloneqq (1) \\ \forall n \in \mathbb{N} \quad \alpha(n+1) \coloneqq \big(\varphi(n)\big)(9) \end{cases} \qquad \begin{cases} \varphi(0) \coloneqq \alpha(0) + (5) = (1) + (5) \\ \forall n \in \mathbb{N} \quad \varphi(n+1) \coloneqq \big(\alpha(n)\big)(7) + (1) \end{cases}$$

My Conjectures :

(1) $\forall n \in \mathbb{N}$ $Sm(\alpha, n)$

$$= \left(\frac{71}{62}\right)\left(\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor + 1} - 1}{62}\right)\right) + \left(\frac{3357}{62}\right)\left(\left(\frac{(63)^{\left\lfloor \frac{(n-1)}{2} \right\rfloor + 1} - 1}{62}\right)\right) - \left(\frac{9}{62}\right)(n+1)$$

(2) $\forall n \in \mathbb{N}$ $Sm(\varphi, n)$

$$= \left(\frac{373}{62}\right)\left(\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor + 1} - 1}{62}\right)\right) + \left(\frac{497}{62}\right)\left(\left(\frac{(63)^{\left\lfloor \frac{(n-1)}{2} \right\rfloor + 1} - 1}{62}\right)\right) - \left(\frac{1}{62}\right)(n+1)$$

2b.i.0

$$\frac{\forall p \in \mathbb{N}(\mathrm{Evn}(p))}{\alpha(p) = (63)^{\left\lfloor \frac{p}{2} \right\rfloor}((1)\left(\frac{1+(-1)^{p}}{2}\right) + (54)\left(\frac{1+(-1)^{p+1}}{2}\right)) + (9)\left(\frac{(63)^{\left\lfloor \frac{p}{2} \right\rfloor}-1}{62}\right)}$$

$\alpha(p)$

$= (63)^{\frac{p}{2}}((1)(1) + (54)(0)) + (9)\left(\frac{(63)^{\frac{p}{2}}-1}{62}\right)$

$= (63)^{\frac{p}{2}}(1) + \left(\frac{9}{62}\right)(63)^{\frac{p}{2}} - \left(\frac{9}{62}\right)$

$\sum^{\left\lfloor \frac{n}{2} \right\rfloor} \alpha(2m)$

$= \sum^{\left\lfloor \frac{n}{2} \right\rfloor}((63)^{m}(1) + (9)\left(\frac{(63)^{m}-1}{62}\right))$

$= (1)\sum^{\left\lfloor \frac{n}{2} \right\rfloor}(63)^{m} + \left(\frac{9}{62}\right)\left(\sum^{\left\lfloor \frac{n}{2} \right\rfloor}(63)^{m} - \left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right)\right)$

$= \left(\frac{71}{62}\right)\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor+1}-1}{62}\right) - \left(\frac{9}{62}\right)\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right)$

$$\frac{\forall q \in \mathbb{N}(\mathrm{Odd}(q))}{\alpha(q) = (63)^{\left\lfloor \frac{q}{2} \right\rfloor}((1)\left(\frac{1+(-1)^{q}}{2}\right) + (54)\left(\frac{1+(-1)^{q+1}}{2}\right)) + (9)\left(\frac{(63)^{\left\lfloor \frac{q}{2} \right\rfloor}-1}{62}\right)}$$

$\alpha(q)$

$= (63)^{\frac{q-1}{2}}((1)(0) + (54)(1)) + (9)\left(\frac{(63)^{\frac{q-1}{2}}-1}{62}\right)$

$= (63)^{\frac{q-1}{2}}(54) + \left(\frac{9}{62}\right)(63)^{\frac{q-1}{2}} - \left(\frac{9}{62}\right)$

$\sum^{\left\lfloor \frac{n-1}{2} \right\rfloor} \alpha(2m+1)$

$= \sum^{\left\lfloor \frac{n-1}{2} \right\rfloor}((63)^{m}(54) + (9)\left(\frac{(63)^{m}-1}{62}\right))$

$= (54)\sum^{\left\lfloor \frac{n-1}{2} \right\rfloor}(63)^{m} + \left(\frac{9}{62}\right)\left(\sum^{\left\lfloor \frac{n-1}{2} \right\rfloor}(63)^{m} - \left(\left\lfloor \frac{n-1}{2} \right\rfloor + 1\right)\right)$

$= \left(\frac{3357}{62}\right)\left(\frac{(63)^{\left\lfloor \frac{n-1}{2} \right\rfloor+1}-1}{62}\right) - \left(\frac{9}{62}\right)\left(\left\lfloor \frac{n-1}{2} \right\rfloor + 1\right)$

$Sm(\alpha,n)$

$= \left(\frac{71}{62}\right)\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor+1}-1}{62}\right) + \left(\frac{3357}{62}\right)\left(\frac{(63)^{\left\lfloor \frac{n-1}{2} \right\rfloor+1}-1}{62}\right) - \left(\frac{9}{62}\right)\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right) - \left(\frac{9}{62}\right)\left(\left\lfloor \frac{n-1}{2} \right\rfloor + 1\right)$

$= \left(\frac{71}{62}\right)\left(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor+1}-1}{62}\right) + \left(\frac{3357}{62}\right)\left(\frac{(63)^{\left\lfloor \frac{n-1}{2} \right\rfloor+1}-1}{62}\right) - \left(\frac{9}{62}\right)(n+1)$

2b.ii.0

$$\frac{\forall p \in \mathbb{N}(\mathrm{Evn}(p))}{\varphi(p) = (63)^{\left\lfloor \frac{p}{2} \right\rfloor}((6)\,(\frac{1+(-1)^{p}}{2}) + (8)\,(\frac{1+(-1)^{p+1}}{2})) + (1)\,(\frac{(63)^{\left\lfloor \frac{p}{2} \right\rfloor}-1}{62})}$$

$\varphi(p)$

$= (63)^{\frac{p}{2}}((6)(1) + (8)(0)) + (1)\,(\frac{(63)^{\frac{p}{2}}-1}{62})$

$= (63)^{\frac{p}{2}}(6) + (\frac{1}{62})\,(63)^{\frac{p}{2}} - (\frac{1}{62})$

$\sum^{\left\lfloor \frac{n}{2} \right\rfloor} \varphi(2m)$

$= \sum^{\left\lfloor \frac{n}{2} \right\rfloor}((63)^{m}(6) + (1)\,(\frac{(63)^{m}-1}{62}))$

$= (6)\sum^{\left\lfloor \frac{n}{2} \right\rfloor}(63)^{m} + (\frac{1}{62})(\sum^{\left\lfloor \frac{n}{2} \right\rfloor}(63)^{m} - (\left\lfloor \frac{n}{2} \right\rfloor+1))$

$= (\frac{373}{62})(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor+1}-1}{62}) - (\frac{1}{62})(\left\lfloor \frac{n}{2} \right\rfloor+1)$

$$\frac{\forall q \in \mathbb{N}(\mathrm{Odd}(q))}{\varphi(q) = (63)^{\left\lfloor \frac{q}{2} \right\rfloor}((6)\,(\frac{1+(-1)^{q}}{2}) + (8)\,(\frac{1+(-1)^{q+1}}{2})) + (1)\,(\frac{(63)^{\left\lfloor \frac{q}{2} \right\rfloor}-1}{62})}$$

$\varphi(q)$

$= (63)^{\frac{q-1}{2}}((6)(0) + (8)(1)) + (1)\,(\frac{(63)^{\frac{q-1}{2}}-1}{62})$

$= (63)^{\frac{q-1}{2}}(8) + (\frac{1}{62})\,(63)^{\frac{q-1}{2}} - (\frac{1}{62})$

$\sum^{\left\lfloor \frac{n-1}{2} \right\rfloor} \varphi(2m+1)$

$= \sum^{\left\lfloor \frac{n-1}{2} \right\rfloor}((63)^{m}(8) + (1)\,(\frac{(63)^{m}-1}{62}))$

$= (8)\sum^{\left\lfloor \frac{n-1}{2} \right\rfloor}(63)^{m} + (\frac{1}{62})(\sum^{\left\lfloor \frac{n-1}{2} \right\rfloor}(63)^{m} - (\left\lfloor \frac{n-1}{2} \right\rfloor+1))$

$= (\frac{497}{62})(\frac{(63)^{\left\lfloor \frac{n-1}{2} \right\rfloor+1}-1}{62}) - (\frac{1}{62})(\left\lfloor \frac{n-1}{2} \right\rfloor+1)$

$Sm(\varphi,n)$

$= (\frac{373}{62})(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor+1}-1}{62}) + (\frac{497}{62})(\frac{(63)^{\left\lfloor \frac{n-1}{2} \right\rfloor+1}-1}{62}) - (\frac{1}{62})(\left\lfloor \frac{n}{2} \right\rfloor+1) - (\frac{1}{62})(\left\lfloor \frac{n-1}{2} \right\rfloor+1)$

$= (\frac{373}{62})(\frac{(63)^{\left\lfloor \frac{n}{2} \right\rfloor+1}-1}{62}) + (\frac{497}{62})(\frac{(63)^{\left\lfloor \frac{n-1}{2} \right\rfloor+1}-1}{62}) - (\frac{1}{62})(n+1)$

5 (2c) Prove by induction, starting on this side and continuing on properly numbered pages that follow, your conjectures from (2a) and (2b) are valid.

SPf

$P(n) \coloneqq (\alpha(n)=(63)\left\lfloor\frac{n}{2}\right\rfloor((1)(\frac{1+(-1)^n}{2})+(54)(\frac{1+(-1)^{n+1}}{2}))+(9)(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62}))$

$\wedge\ (\quad \varphi(n)=(63)\left\lfloor\frac{n}{2}\right\rfloor((6)(\frac{1+(-1)^n}{2})+(8)(\frac{1+(-1)^{n+1}}{2}))+(1)(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62}))$

BC

LS $= \alpha(0)=1$

RS $= (63)^0(1)+(9)((63)^0-1)\div 62 = (1)(1)+(9)(0) = 1$

LS = RS

LS $= \varphi(0)=\alpha(0)+5=1+5=6$

RS $= (63)^0(6)+(1)((63)^0-1)\div 62 = (1)(6)+(1)(0) = 6$

LS = RS

$P(0)$

LS $= \alpha(1)=(9)\varphi(0)=(9)(6)=54$

RS $= (63)^0(54)+(9)((63)^0-1)\div 62 = (1)(54)+(9)(0) = 54$

LS = RS

LS $= \varphi(1)=(7)\alpha(0)+1=(7)(1)+1=8$

RS $= (63)^0(8)+(1)((63)^0-1)\div 62 = (1)(8)+(1)(0) = 8$

LS = RS

$P(1)$

RcS

$P(0)\ P(1)\ \cdots\ P(n)$

LS

$= \alpha(n+2)$

$= (\alpha(n)(7)+1)(9)$

$= (63)\alpha(n)+9$

$= (63)((63)\left\lfloor\frac{n}{2}\right\rfloor((1)(\frac{1+(-1)^n}{2})+(54)(\frac{1+(-1)^{n+1}}{2}))+(9)(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62}))+9$

$=$ RS

LS

$= \varphi(n+2)$

$= (\varphi(n)(9))(7)+1$

$= (63)\varphi(n)+1$

$= (63)((63)\left\lfloor\frac{n}{2}\right\rfloor((6)(\frac{1+(-1)^n}{2})+(8)(\frac{1+(-1)^{n+1}}{2}))+(1)(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62}))+1$

$=$ RS

$P(n+2)$

$P(0)\wedge P(1)\wedge\forall n\in\mathbb{N}(P(n)\rightarrow P(n+2))\rightarrow\forall n\in\mathbb{N}(P(n))$

EPf

(2c) continued

SPf

$R(n) \coloneqq (Sm(\alpha,n)=(\frac{71}{62})(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62})+(\frac{3357}{62})(\frac{(63)\left\lfloor\frac{(n-1)}{2}\right\rfloor-1}{62})-(\frac{9}{62})(n+1))$

$\wedge\,(\quad Sm(\varphi,n)=(\frac{373}{62})(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62})+(\frac{497}{62})(\frac{(63)\left\lfloor\frac{(n-1)}{2}\right\rfloor-1}{62})-(\frac{1}{62})(n+1))$

BC

LS $= Sm(\alpha,0) \coloneqq \alpha(0)=1$

RS $= (\frac{71}{62})(1)+(\frac{3357}{62})(0)-(\frac{9}{62})(1) = 1$

LS = RS

LS $= Sm(\varphi,0) \coloneqq \varphi(0)=6$

RS $= (\frac{373}{62})(1)+(\frac{497}{62})(0)-(\frac{1}{62})(1) = 6$

LS = RS

$R(0)$

RcS

$R(0)\ R(1)\ \cdots\ R(n)$

LS

$= Sm(\alpha,n+1)$

$= Sm(\alpha,n)+\alpha(n+1)$

$= (\frac{71}{62})(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62})+(\frac{3357}{62})(\frac{(63)\left\lfloor\frac{(n-1)}{2}\right\rfloor-1}{62})-(\frac{9}{62})(n+1)+\left((63)\left\lfloor\frac{n+1}{2}\right\rfloor((1)(\frac{1+(-1)^{n+1}}{2})+(54)(\frac{1+(-1)^{n+1+1}}{2}))+(9)(\frac{(63)\left\lfloor\frac{n+1}{2}\right\rfloor-1}{62})\right)$

$=$ RS

LS

$= Sm(\varphi,n+1)$

$= Sm(\varphi,n)+\varphi(n+1)$

$= (\frac{373}{62})(\frac{(63)\left\lfloor\frac{n}{2}\right\rfloor-1}{62})+(\frac{497}{62})(\frac{(63)\left\lfloor\frac{(n-1)}{2}\right\rfloor-1}{62})-(\frac{1}{62})(n+1)+\left((63)\left\lfloor\frac{n+1}{2}\right\rfloor((6)(\frac{1+(-1)^{n+1}}{2})+(8)(\frac{1+(-1)^{n+1+1}}{2}))+(1)(\frac{(63)\left\lfloor\frac{n+1}{2}\right\rfloor-1}{62})\right)$

$=$ RS

$R(n+1)$

$R(0) \wedge \forall n \in \mathbb{N}(R(n) \rightarrow R(n+1)) \rightarrow \forall n \in \mathbb{N}(R(n))$

EPf

6 (3a) Consider the following recurrence relation for $\varphi(n)$ and **answer the questions**:

$$\begin{cases} \varphi(0) \coloneqq c \\ \forall n \in \mathbb{N} \qquad \varphi(n+1) \coloneqq \big(\varphi(n)\big)b + d \end{cases}$$

Show calculations on pages that follow, numbered according to instructions.

Rewrite the recursion using your values of $a, b, c,$ and d.

$$\begin{cases} \varphi(0) \coloneqq (\,5\,) \\ \forall n \in \mathbb{N} \qquad \varphi(n+1) \coloneqq \big(\varphi(n)\big)(\,9\,) + (\,7\,) \end{cases}$$

Show computations on pages that follow, numbered as per instructions, for making a conjecture:

2 (i) Conjecture:

$$\forall n \in \mathbb{N} \qquad \varphi(n) = (5)\,(9)^n + (7)\left(\frac{9^n - 1}{8}\right)$$

4 (i) Prove your conjecture by induction:

SPf

$P(n) \coloneqq (\varphi(n) = (5)(9)^n + (7)(\frac{9^n - 1}{8}))$

BC

LS $= \varphi(0) = 5$

RS $= (5)(1) + (7)(\frac{0}{8}) = 5$

LS = RS

$P(0)$

RcS

$P(0)\ P(1) \cdots P(n)$

LS

$= \varphi(n+1)$

$= \varphi(n)(9)+7$

$= ((5)(9)^n + (7)(\frac{9^n - 1}{8}))(9)+7$

$= (5)(9)^{n+1} + (63)(\frac{9^n - 1}{8}) + 7$

$= (5)(9)^{n+1} + (7)(\frac{9^{n+1} - 1}{8})$

$=$ RS

$P(n+1)$

$P(0)$,

$P(0)\ P(1) \cdots P(n)$

$P(n+1)$

$\forall n \in \mathbb{N}(P(n))$

EPf

3a.0

$$\frac{\varphi(n+1)=\varphi(n)(9)+7}{}\ \langle n \leftarrow 0 \rangle$$

$$\varphi(0+1)=\varphi(0)(9)+7$$

$$\varphi(1)=(5)(9)+7$$

$$=\ 52$$

3a.1

$$\frac{\varphi(n+1)=\varphi(n)(9)+7}{}\ \langle n \leftarrow 1 \rangle$$

$$\varphi(1+1)=\varphi(1)(9)+7$$

$$\varphi(2)=(52)(9)+7$$

$$=\ 475$$

3a.2

$$\frac{\varphi(n+1)=\varphi(n)(9)+7}{}\ \langle n \leftarrow 2 \rangle$$

$$\varphi(2+1)=\varphi(2)(9)+7$$

$$\varphi(3)=(475)(9)+7$$

$$=\ 4282$$

3a.3

$$\varphi(n)$$

$$=\ 9\varphi(n-1)+7$$

$$=\ 9(9\varphi(n-2)+7)+7$$

$$=\ 9^{2}\varphi(n-2)+(9)(7)+7$$

$$=\ 9^{2}(9\varphi(n-3)+7)+(9)(7)+7$$

$$=\ 9^{3}\varphi(n-3)+9^{2}(7)+(9)(7)+7$$

$$=\ 9^{n}\varphi(0)+7(9^{n-1}+9^{n-2}+\cdots+9+1)$$

3a.4

$$9^{n-1}+\cdots+9+1$$

$$=\ (9^{n}-1)\div(9-1)$$

$$=\ (9^{n}-1)\div 8$$

3a.5

$$\varphi(n)$$

$$=\ 9^{n}(5)+7((9^{n}-1)\div 8)$$

$$=\ (5)(9)^{n}+(7)((9^{n}-1)\div 8)$$

6 (3b) Prove, following the method shown in class, that: $(abcd)^{\frac{1}{abcd+1}}$ **is irrational**, that is: $\cancel{Rtl}\left((abcd)^{\frac{1}{abcd+1}}\right)$.

First rewrite the problem using your values of a, b, c and d.

My Problem:

$$\cancel{Rtl}\left(((1\ )(9\ )(5\ )(7\ ))^{\frac{1}{(1\ )(9\ )(5\ )(7\ )+1}}\right)$$

that is: $\cancel{Rtl}\left((_{315})^{\frac{1}{(_{316})}}\right)$

Start proof on this side and continue on pages that follow, numbered according to instructions.

SPf

$$\frac{\neg(\cancel{Rtl}((\cancel{315})^{\frac{1}{316}}))}{Rtl((315)^{\frac{1}{316}})}$$

$$\frac{Rtl((315)^{\frac{1}{316}})}{\text{for some } p,q\in\mathbb{N}\ ((\gcd(p,q)=1)\wedge((315)^{\frac{1}{316}}=p\div q))}$$

$$\frac{\text{for some } p,q\in\mathbb{N}\ ((\gcd(p,q)=1)\wedge((315)^{\frac{1}{316}}=p\div q))}{(\gcd(p,q)=1)\qquad((315)^{\frac{1}{316}}=p\div q)}$$

$$\frac{(\gcd(p,q)=1)\qquad((315)^{\frac{1}{316}}=p\div q)}{(315)^{\frac{1}{316}}=p\div q}$$

(3b) continued - proof that $(315)^{\frac{1}{316}}$ is irrational

$$((315)^{\frac{1}{316}})^{316} = (p \div q)^{316}$$

$$315 = p^{316}$$

$$315 = p^{316} \div q^{316}$$

$$315q^{316} = p^{316}$$

$$(3)(3)(5)(7)(q^{316}) = p^{316}$$

$$\text{prm}(3) \qquad 3 | p^{316}$$

$$3 | p$$

$$\text{for some } r \in \mathbb{N} \ (p = 3r)$$

$$315q^{316} = (3r)^{316}$$

$$315q^{316} = 3^{316} r^{316}$$

$$35q^{316} = 3^{314} r^{316} \quad (\div 9)$$

$$\text{prm}(3) \qquad 3 | 35q^{316}$$

$$3 | q^{316} \quad (\gcd(3,35) = 1)$$

$$3 | q$$

3b.i

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{3|p \qquad 3|q}{3|\gcd(p,q)}}{3|\gcd(p,q) \qquad \gcd(p,q)=1}}{3|1}}{\bot}}{\neg(\neg(\cancel{Rtl((315)\ \frac{1}{316})}))}$$

$$\cancel{Rtl((315)\ \frac{1}{316})}$$

EPf

2 (3c1) Consider the numbers defined below, in which the dot indicates a decimal point, and the string in parentheses is the repeating part of a repeating decimal.

$$x := .(bc) \qquad = \qquad .\left((9)(5)\right)$$

Show **computations on pages that follow, numbered according to instructions,** and record answers on this side.

1 (i) Find a recursive definition for the **digits** of x after the decimal point, with the recursion starting at $n = 0$ and prove inductively that the definition is correct.

BC $x_0 := 9,\ x_1 := 5$
RcS $\forall n \in \mathbb{N}\ x_{n+2} := x_n$

1 (i) Prove that: $Rtl(x)$**,** that is, **x is rational**.

(3c1) continued

SPf

$P(n) \coloneqq (x_n = 9(\frac{1+(-1)^n}{2}) + 5(\frac{1+(-1)^{n+1}}{2}))$

BC

LS $= x_0 = 9$

RS $= 9(\frac{1+(-1)^0}{2}) + 5(\frac{1+(-1)^1}{2}) = 9(1)+5(0) = 9$

LS = RS

$P(0)$

LS $= x_1 = 5$

RS $= 9(\frac{1+(-1)^1}{2}) + 5(\frac{1+(-1)^2}{2}) = 9(0)+5(1) = 5$

LS = RS

$P(1)$

RcS

$P(0)\ P(1) \cdots\ P(n)$

LS

$= x_{n+2}$

$= x_n$

$= 9(\frac{1+(-1)^n}{2}) + 5(\frac{1+(-1)^{n+1}}{2})$

$(-1)^{n+2} = (-1)^n((-1)^2) = (-1)^n$

$= 9(\frac{1+(-1)^{n+2}}{2}) + 5(\frac{1+(-1)^{n+3}}{2})$

$=$ RS

$P(n+2)$

$P(0) \wedge P(1) \wedge \forall n \in \mathbb{N}(P(n) \rightarrow P(n+2)) \rightarrow \forall n \in \mathbb{N}(P(n))$

EPf

SPf

$x = .959595...$

$(100)x = 95.959595...$

$(100)x - x = 95$

$(99)x = 95$

$x = \frac{95}{99}$

$95,99 \in \mathbb{N}$, $99 \neq 0$, $\gcd(95,99)=1$

$Rtl(x)$

EPf

3 (3c2) Consider the numbers defined below, in which the dot indicates a decimal point.

$$y := .ddcdccdcccd\ldots \qquad = 0.77575575557\ldots$$

where for every $\forall n \in \mathbb{N}$ there are n c's between the nth and the $(n+1)^{st}$ d.

Show **computations on pages that follow, numbered according to instructions,** and record answers on this side.

1 (iii) Find a recursive definition for the **digits** of y after the decimal point, with the recursion starting at $n = 0$ and prove inductively that the definition is correct.

BC $p(0) := 0$
RcS $\forall n \in \mathbb{N}\ p(n+1) := p(n) + (n+1)$

2 (iv) Prove that: ~~Rtl~~(y), that is, **y is irrational**.

(3c2) continued

SPf

$P(n) \coloneqq (p(n) = (\frac{n(n+1)}{2}))$

BC

LS $= p(0) = 0$

RS $= (0)(1) \div 2 = 0$

LS = RS

$P(0)$

RcS

$P(0)\ P(1) \cdots\ P(n)$

LS

$= p(n+1)$

$= p(n)+(n+1)$

$= \frac{n(n+1)}{2}+(n+1)$

$= \frac{n(n+1)+2(n+1)}{2}$

$= \frac{(n+1)(n+2)}{2}$

$=$ RS

$P(n+1)$

$P(0) \wedge \forall n \in \mathbb{N}(P(n) \rightarrow P(n+1)) \rightarrow \forall n \in \mathbb{N}(P(n))$

EPf

SPf

$$\frac{Rtl(y)}{\frac{\text{for some } m \in \mathbb{N}\ (m>0)\ \ \text{for some } l_0 \in \mathbb{N}}{\frac{\forall k \geq l_0\ (\text{dig}(k)=\text{dig}(k+m))}{\frac{\text{for some } B \in \mathbb{N}\ \forall\ n \in \mathbb{N}\ (p(n+1)-p(n) \leq B)}{\frac{p(n+1)-p(n) = n+1}{\frac{\text{for some } B \in \mathbb{N}\ \forall\ n \in \mathbb{N}\ (n+1 \leq B)}{\frac{\bot}{\frac{\neg Rtl(y)}{\cancel{Rtl(y)}}}}}}}}}$$

EPf